

Watch Item: HED-Family Pairing within Q1 2026 Fireball Activity

Status	Open watch item
Scope	Composition-only, tracked separately from directional/fireball-rate baseline

1. Observed Facts

- Q1 2026 shows elevated upper-tail fireball activity, not a clear increase in total counts.
- ~2,300 total events
- 40 events with 50+ reports (approximately 2x typical)
- 16 events with 100+ reports (approximately 2x typical)
- 33/40 large events associated with sonic booms

Two distinct radiant concentrations observed:

- Anthelion zone
- High-declination zone

Two meteorite-producing events within 9 days:

- March 8 (Western Europe) -> suspected diogenite
- March 17 (Ohio) -> confirmed eucrite

Angular separation between events:

- ~98.2 degrees -> not from a common stream

Historical baseline (2015-2025 Meteoritical Bulletin):

- 116 classified falls
- 7 HED-classified falls (~6.0%)

2. Baseline Explanation (Directional)

The working baseline for Q1 2026 is:

Directional geometry (Anthelion + high-declination) increases the rate of slow, recoverable events

This explains:

- why more large fireballs are observed
- why more meteorites are recoverable

It does not explain:

- compositional clustering (e.g., HED pairing)

3. Anomaly Under Watch (Compositional)

Observed anomaly:

Two HED-family meteorite falls within 9 days

Scope:

- not a shared-stream claim
- not a global explanation
- purely a composition coincidence under scrutiny

4. Probability Framing

Two distinct probability statements:

A. Conditional (given two classified falls)

Probability both are HED:

$$p^2 \approx (0.06)^2 \approx 0.0036$$

Approximately 0.36% (~1 in 275)

Note: Measures composition rarity only

B. Post-hoc (over full 2015-2025 record)

Probability at least one 9-day HED-HED pairing appears somewhere:

Approximately 9%

Approximately 4-5% under stricter assumptions

Note: Measures whether such clustering can appear by chance

5. Interpretation

- The pair is unusual
- But not yet statistically decisive

Current classification: watch item, not established clustering

6. Assumptions

- Baseline uses 2015-2025 Meteoritical Bulletin valid falls
- Combines confirmed + probable falls
- Assumes constant HED rate (~6%)
- Treats March 8 event as HED-family (pending confirmation)
- Uses uniform-in-time null model

7. Known Limitations

- Small sample size (7 historical HED falls)
- One event classification not yet confirmed
- Null model does not include:
 - source-specific composition variation
 - recovery bias
 - directional filtering
- Radiant uncertainty: 10-20 degrees typical error
- Possible reporting effects: witness amplification not fully quantified

8. Strengthening Conditions

The anomaly becomes stronger if:

- March 8 fall is confirmed as HED
- additional near-term HED falls occur
- HED fraction rises above baseline (~6%)

9. Weakening / Falsification

The anomaly weakens if:

- March 8 event is not HED
- HED fraction remains near baseline
- no further clustering appears

This is a cumulative test, not a single-event test

10. Tracking Protocol

Use future classified falls as denominator.

Detection thresholds:

- 4+ HED in next 20 falls -> early signal
- 6+ HED in next 43 falls -> strong signal (~tripling)
- 14+ HED in next 140 falls -> moderate signal (~doubling)

Negative indicators:

- ≤ 3 HED in next 40 falls -> weakens strong clustering
- ≤ 2 HED in next 60 falls -> weakens moderate clustering

11. Current Read

Q1 2026 appears to be an upper-tail fireball anomaly driven largely by directional geometry (Anthelion + high-declination)

The HED pairing is a separate compositional anomaly that is unusual but not yet statistically decisive

12. Operational Summary

- Treat as watch item
- Do not merge with:
 - directional explanations
 - comet activity
 - unrelated compositional anomalies
- Update only when:
 - new classified falls are added
 - classifications are confirmed or revised

13. Guardrail

Do not convert this into a narrative.
Do not merge independent signals without evidence.
Do not increase certainty beyond what the data supports.